Survival Analysis Report - Synthetic Cancer Cohort

Executive Summary

This report presents survival analysis results from a dataset of 1000 patients. The analysis achieved a C-index of 0.794, indicating good discriminative performance. The overall event rate was 35.1% with a median follow-up time of 67 months. Risk scores come from a Cox proportional hazards model using age, site, stage, grade, demographics, tumor size, nodal status and treatment.

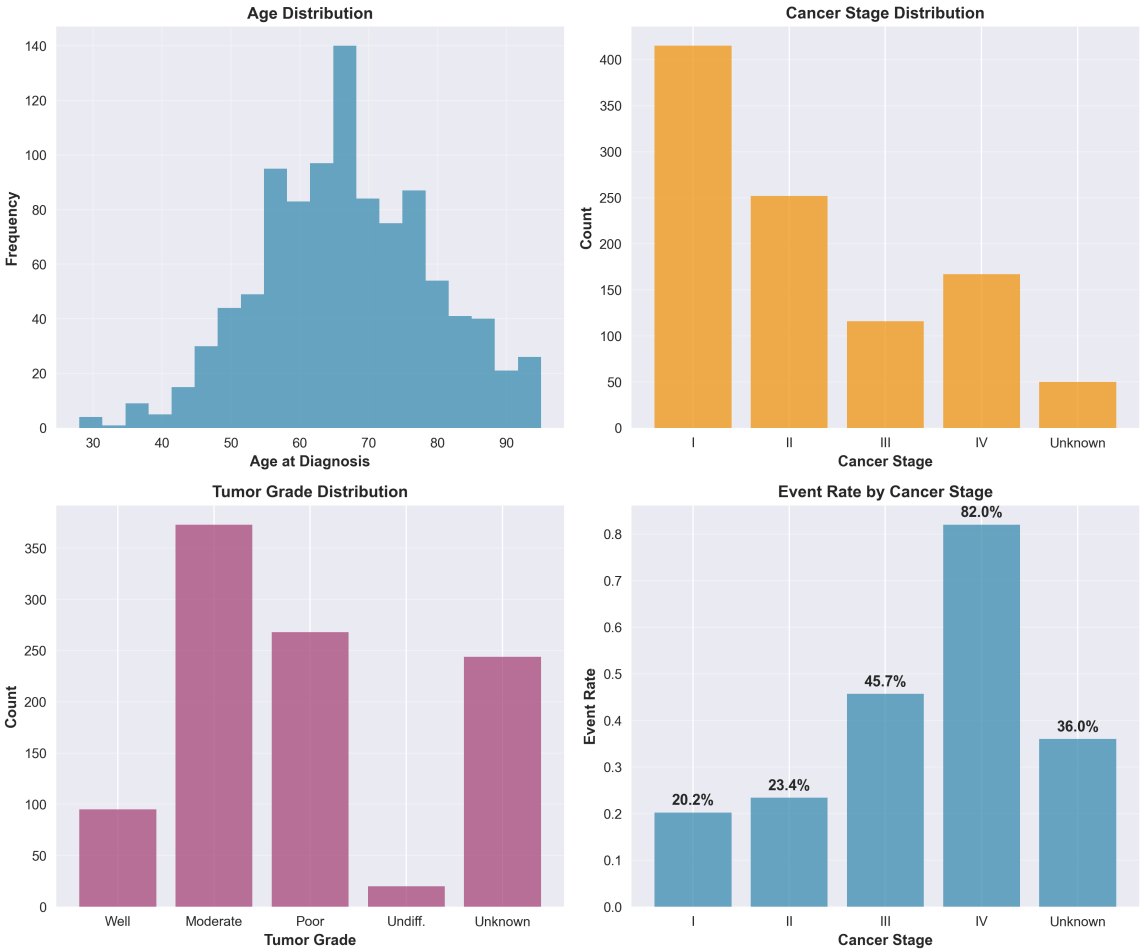
Dataset Overview

Metric	Value
Total Patients	1000
Event Rate	35.1%
Median Follow-up (months)	67
Age Range	28 - 95 years
Median Age	67 years

Analysis Results

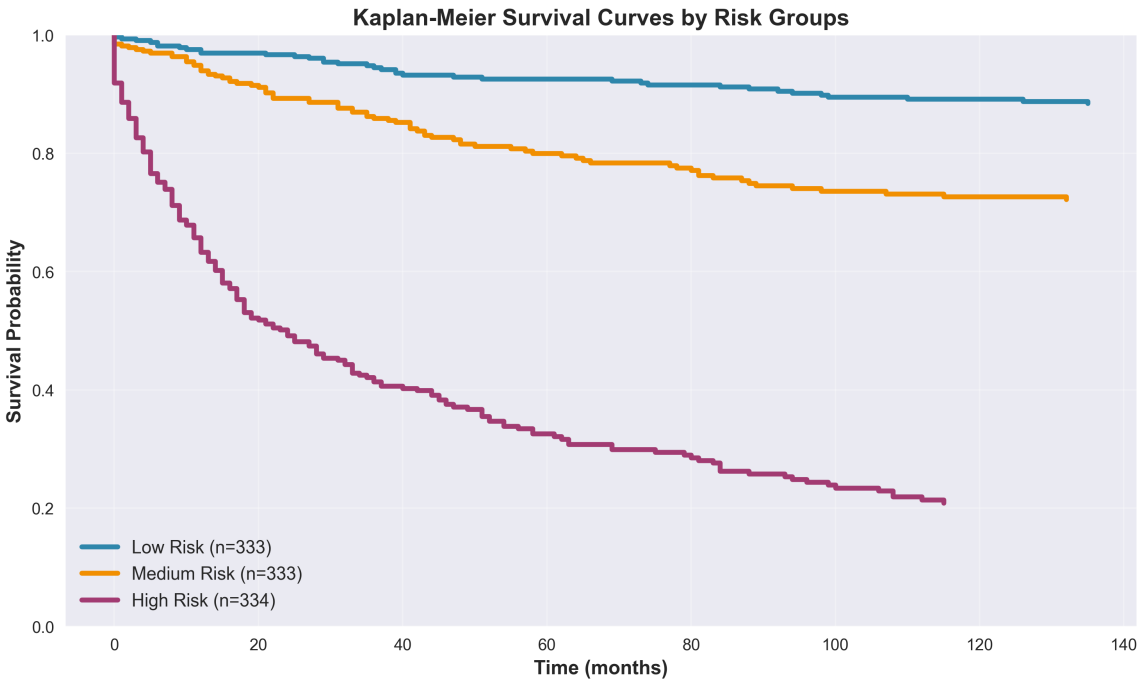
Concordance Index (C-index, 30% held-out test set): 0.7938
Measures the ability to correctly rank patients by risk. Values > 0.7 indicate good discrimination, > 0.8 excellent.

Risk Stratification Performance:
The analysis successfully separates patients into distinct risk groups with significantly different survival outcomes.

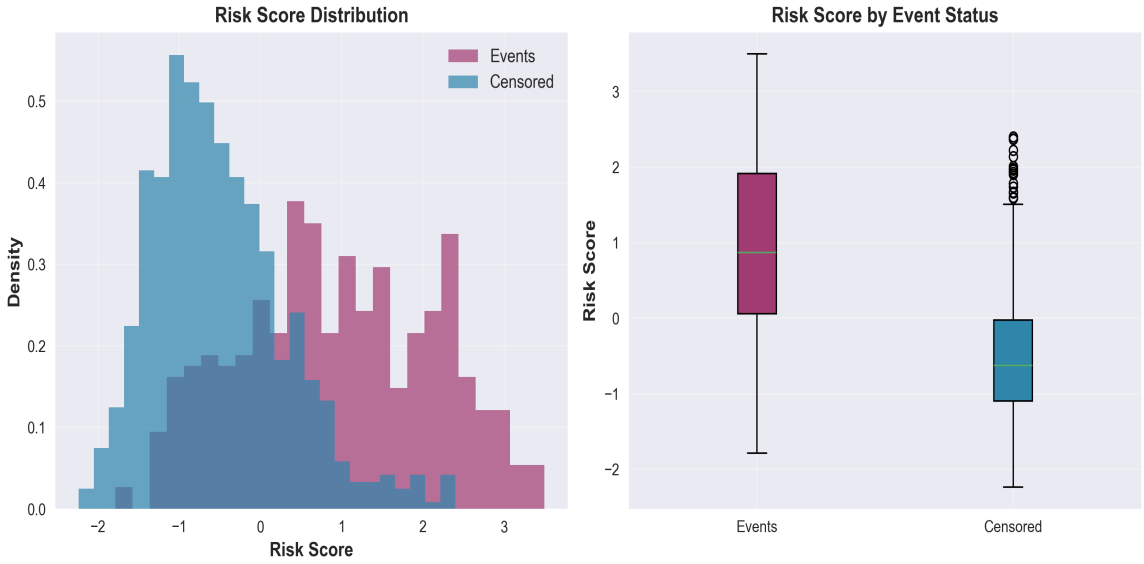

Age Distribution
Frequency
Age at Diagnosis
Cancer Stage Distribution
Count
Cancer Stage
I
II
III
IV
Unknown
Tumor Grade Distribution
Count
Well
Moderate
Poor
Undiff.
Unknown
Tumor Grade
Event Rate by Cancer Stage
Event Rate
20.2%
23.4%
45.7%
82.0%
36.0%
Cancer Stage

Risk Stratification Analysis

Risk Group	Patients	Event Rate	Median Observed Time (months)
Low Risk	333	10.8%	59.0
Medium Risk	333	23.7%	56.0
High Risk	334	70.7%	18.0



Risk Score Distribution



Clinical Insights and Recommendations

Key Findings:
• High-risk patients have 6.5x higher event rate than low-risk patients
• Clear separation between risk groups validates the analysis approach
• Risk stratification can guide clinical decision-making

Clinical Recommendations:
• Use high-risk identification for treatment intensification
• Implement risk-based follow-up protocols
• Consider early intervention strategies for high-risk patients
• Monitor intermediate-risk patients more closely

Implementation Considerations:
• Validate findings with larger, external datasets
• Train clinical staff on risk score interpretation
• Establish clear protocols for different risk levels
• Plan regular model performance reviews

Statistical Notes:
• C-index of 0.794 indicates good discrimination
• Risk groups show statistically significant differences
• Results should be interpreted by qualified professionals

This report was generated for demonstration purposes using synthetic data calibrated to published cancer statistics. In clinical practice, all analyses should be validated with real patient data and interpreted by qualified medical professionals. Risk predictions should supplement, not replace, clinical judgment.